

Table 1: Measured lengths and Lorentz force with  $I_{\text{obs}} = 3.04 \text{ A}$

| 迴路板編號 | outer length (mm) | inner length (mm) | average length (mm) | force (gw) | force (N) |
|-------|-------------------|-------------------|---------------------|------------|-----------|
| 37    | 21.8              | 18.0              | 19.9                | 0.56       | 0.00549   |
| 38    | 42.0              | 38.0              | 40.0                | 1.06       | 0.0104    |
| 38-2  | 42.0              | 34.0              | 38.0                | 0.98       | 0.00961   |
| 39    | 31.9              | 28.1              | 30.0                | 0.82       | 0.00804   |
| 40    | 12.0              | 8.0               | 10.0                | 0.18       | 0.00177   |

Table 2: Linear fit results with  $B_{\text{obs}} = 96.13 \text{ mT}$

| parameter                          | outer                           | average                         | inner                           |
|------------------------------------|---------------------------------|---------------------------------|---------------------------------|
| slope $s$                          | $0.265 \pm 0.025$               | $0.276 \pm 0.020$               | $0.285 \pm 0.017$               |
| intercept                          | $(-8.6 \pm 8.1) \times 10^{-4}$ | $(-5.4 \pm 5.8) \times 10^{-4}$ | $(-1.3 \pm 4.7) \times 10^{-4}$ |
| $B = s/I_{\text{obs}} \text{ (T)}$ | $0.0870 \pm 0.0083$             | $0.0907 \pm 0.0064$             | $0.0938 \pm 0.0056$             |
| error                              | -9.47%                          | -5.68%                          | -2.42%                          |

Table 3: 改變磁鐵數量，測量磁場與洛倫茲力的大小。使用 38 號板

| magnet number | $B \text{ (mT)}$ | force (gw) | force (N) |
|---------------|------------------|------------|-----------|
| 8             | 97.51            | 1.12       | 0.01098   |
| 7             | 84.58            | 1.00       | 0.009807  |
| 6             | 73.51            | 0.86       | 0.008434  |
| 5             | 61.12            | 0.72       | 0.007061  |
| 4             | 46.58            | 0.58       | 0.005688  |
| 3             | 36.27            | 0.44       | 0.004315  |
| 2             | 22.03            | 0.28       | 0.002746  |
| 1             | 13.32            | 0.12       | 0.001177  |
| 0             | 0                | 0.00       | 0         |

Table 4: 線性擬合的結果。  $L = 38.0 \text{ mm}$  是 38 號板的內徑，

| parameter                    | value                       |
|------------------------------|-----------------------------|
| slope $s$                    | $0.1146 \pm 0.0024$         |
| intercept                    | $(4 \pm 14) \times 10^{-5}$ |
| $I = s/L \text{ (A)}$        | $3.015 \pm 0.063$           |
| $I_{\text{obs}} \text{ (A)}$ | 3.04                        |
| error                        | -0.84%                      |

Table 5: Measured mass and calculated force versus power-supply current

| mass (g) | force ( $10^{-4} \text{ N}$ ) | current (A) |
|----------|-------------------------------|-------------|
| 0.0349   | 3.423                         | 5.509       |
| 0.0474   | 4.648                         | 6.410       |
| 0.0524   | 5.139                         | 6.810       |
| 0.0647   | 6.345                         | 7.590       |
| 0.0756   | 7.414                         | 8.111       |

Table 6: Derived parameters from the linear fit and measurement

| parameter                                             | value                              |
|-------------------------------------------------------|------------------------------------|
| slope $s$                                             | $(1.109 \pm 0.028) \times 10^{-5}$ |
| intercept                                             | $(4 \pm 14) \times 10^{-6}$        |
| 反射鏡到動導線的距離 $a \text{ (cm)}$                           | 21.15                              |
| 反射鏡到標尺的距離 $b \text{ (cm)}$                            | 100.65                             |
| 光點位移 $D \text{ (cm)}$                                 | 1.77                               |
| 動導線直徑 $2r_0 \text{ (mm)}$                             | 3.10                               |
| 兩導線中心的距離 $d = 2r_0 + aD/2b \text{ (m)}$               | 0.00496                            |
| 動導線長度 $L \text{ (cm)}$                                | 27.90                              |
| $\mu_{\text{est}} = 2\pi sd/L \text{ (N/A}^2\text{)}$ | $(1.239 \pm 0.032) \times 10^{-6}$ |
| $\mu_0 \text{ (N/A}^2\text{)}$                        | $1.257 \times 10^{-6}$             |
| error                                                 | -1.44%                             |

Table 7: Predicted mass from fitted model and comparison with measured mass

| current (A) | force (N)                          | predicted mass (kg)                | measured mass (kg) | error  |
|-------------|------------------------------------|------------------------------------|--------------------|--------|
| 8.211       | $(7.520 \pm 0.064) \times 10^{-4}$ | $(7.669 \pm 0.065) \times 10^{-5}$ | $7.73e - 05$       | -0.79% |